

Since $H^2(X^{\text{an}}, \mathcal{O}_{X^{\text{an}}})$ is torsion free, any torsion element of $H^2(X; \mathbb{Z})$ vanishes under β . Therefore, to conclude the proof, it suffices to show α is an injection. Since $H^1(X, \mathcal{O}_{X^{\text{an}}}^\times)$ is identified with the Picard group, to prove the desired injection, we only need to show $H^1(X^{\text{an}}, \mathcal{O}_{X^{\text{an}}}) = 0$. Using GAGA for Deligne-Mumford stacks, [Hal11, Proposition A.4], we have $H^1(X^{\text{an}}, \mathcal{O}_{X^{\text{an}}}) = H^1(X, \mathcal{O}_X)$. Since $H^1(X; \mathbb{Q}) = 0$, we also have $H^1(X; \mathbb{C}) = 0$, and hence we conclude $H^1(X, \mathcal{O}_X) = 0$ using [Sat12, Corollary 1.7], which says that the Hodge de Rham spectral sequence degenerates for smooth proper Deligne-Mumford stacks. $\square$

7.3. Proving the stable Picard rank conjecture. We now aim to prove Theorem 7.1.1. To do this, we next compute the first two stable cohomology groups of $[[\text{CHur}_{\mathbb{P}^1, n}^{G, c} / G] / \text{PGL}_2]$. To do so, we need a basic lemma about the number of connected components of Hurwitz spaces.

Lemma 7.3.1. *Let G be a group and $c \subset G$ a conjugacy class generating G . For n sufficiently large, the set of connected components of CHur_n^c with boundary monodromy $g \in G$ is either empty or forms a torsor under $H_2(G, c)$; it is nonempty if and only if the image of n in G^{ab} (under the map $\mathbb{Z} \rightarrow G^{\text{ab}}$ sending the positive generator to the image of any element of c) agrees with the image of g in G^{ab} .*

Rephrasing the statement above, there are $H_2(G, c)$ many components if the image of n in G^{ab} agrees with the image of g , and 0 components otherwise.

Proof. This essentially follows from [Woo21] as we now explain. Indeed, using [Woo21, Theorem 2.5 and Theorem 3.1] we can identify the number of components of CHur_n^c for n sufficiently large with the set of elements in a certain reduced Schur cover $S_c \rightarrow G$ having the same image in G^{ab} as n . Moreover, the boundary monodromy of these components is the same as their image in G under the map $S_c \rightarrow G$. The kernel of $S_c \rightarrow G$ is identified with $H_2(G, c)$ and so connected components with boundary monodromy g either form a torsor under $H_2(G, c)$ when the image of n in G^{ab} agrees with the image of g , or else there are no such connected components. $\square$

For the next lemma and its proof, the reader may wish to recall notation from Notation 2.4.1.

Lemma 7.3.2. *Let G be a group, $c \subset G$ be a conjugacy class generating G , and $R := \mathbb{Z}[1/2|G|]$. For n sufficiently large depending on c and for each component $\tilde{Z} \subset [\text{CHur}_{\mathbb{P}^1, n}^{G, c} / G]$, with corresponding component $Z \subset [[\text{CHur}_{\mathbb{P}^1, n}^{G, c} / G] / \text{PGL}_2]$, we have*

$$\begin{aligned} H^1(\tilde{Z}; R) &= H^1(Z; R) = 0, \\ H^2(\tilde{Z}; R) &= H^2(Z; R) = ((\mathbb{Z} / (2n - 2)\mathbb{Z}) \otimes R). \end{aligned}$$

Proof. Taking $g = \text{id}$ in Lemma 7.3.1, we obtain that for n sufficiently large, both $[\text{CHur}_n^{G, c, \partial \in \text{id}} / G]$ and $[\text{CHur}_{\mathbb{P}^1, n}^{G, c} / G]$ have $|H_2(G, c)|$ many connected components. Indeed, the statement for $[\text{CHur}_n^{G, c, \partial \in \text{id}} / G]$ follows from Lemma 7.3.1 and the fact that G conjugation acts trivially on $H_2(G, c)$ as it is identified with the central kernel of $S_c \rightarrow G$ by definition. Since $[\text{CHur}_n^{G, c, \partial \in \text{id}} / G]$ is dense open in $[\text{CHur}_{\mathbb{P}^1, n}^{G, c} / G]$, we obtain $[\text{CHur}_{\mathbb{P}^1, n}^{G, c} / G]$ also has